

Abhishek Gupta

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Professional Summary

Machine Learning Engineer with research and hands-on experience developing end-to-end AI systems across Computer Vision, NLP, and Biomedical Signal Processing. Built transformer-based image classification models achieving 99% test accuracy on 11,000+ images, engineered retrieval-augmented generation (RAG) pipelines with hybrid search, and developed interpretable deep-learning solutions using Grad-CAM and Partial-LRP. Published first-author researcher at ICSS-2025 with expertise in PyTorch, model evaluation, explainable AI, and production-oriented AI deployment.

Technical Expertise

ML & Deep Learning:	PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas
Computer Vision & XAI:	ViT, Swin Transformer, Grad-CAM, Partial-LRP
Modeling & Evaluation:	Transfer Learning, Data Augmentation, Cross-Validation
Deployment:	FastAPI, Docker, Streamlit
Tools:	Linux, Git, PySpark, Hadoop

Key Projects

Medical Research Intelligence System (Offline RAG Platform)

Sept – Oct 2025

- Designed a modular three-layer architecture separating document ingestion, retrieval orchestration, and user-facing services to improve maintainability and future scalability.
- Architected an offline, modular RAG system partitioned into three clean directory layers: `apps/api/nlp` (parsing), `apps/api/rag` (inference), and `apps/web` (**Streamlit** UI).
- Built a hybrid retrieval engine coupling **BM25** lexical ranking with dense **MiniLM-L6-v2** semantic embeddings, optimized via **MMR** to enforce non-redundant top- k context selection.
- Programmed a structured text parser (**PyPDF2** + regex) dividing documents into section-weighted chunks (Abstract, Results, Conclusion) with disk-cached sub-second re-querying.
- Integrated **DistilBART-CNN** over chunked batches for long texts, securing an end-to-end CPU execution latency of **5–15 s** at under 1 GB RAM with a 500 MB cache footprint.
- Deployed a responsive web dashboard supporting audience-adaptive output modes (*Expert / Patient*) with inline evidence citation and zero external API dependencies.

Computational Asset Risk Pipeline | Quantitative Volatility Analytics

March – April 2025

- Engineered a GARCH-family comparative analysis pipeline in **Python** processing five-year BTC/USD high-frequency tick data; modelled asymmetric conditional volatility using **GARCH**, **EGARCH** variants to characterise heavy-tail return distributions.
- Structured statistical risk-simulation scripts driven by Student's t -distributions to systematically evaluate heavy-tail vulnerabilities and severe market drawdown risks under stress-test parameter spaces.
- Research outcomes peer-reviewed and presented as first author at the **5th International Conference on Smart Systems (ICSS-2025)**, Manipal University Jaipur — published paper: *“Asymmetry and Heavy Tails in Bitcoin Volatility: A GARCH Comparative Study”*.

Technical Experience

Computer Vision Research Engineer | Central University of Rajasthan

Jun 2025 – Nov 2025

- Designed and executed a complete deep-learning pipeline for **10-class tomato leaf disease classification** on 11,000 labelled images (1,100 per class); applied 224×224 resizing, channel-wise normalisation, and a 400-variant-per-image augmentation strategy yielding an effective **4 M training instances**.
- Benchmarked fine-tuned **Swin Transformer** against **Vision Transformer (ViT-B/16)** under identical 5-fold stratified evaluation: Swin achieved **99% test accuracy** (macro F1 = 0.99) vs. ViT's **94% test accuracy** (macro F1 = 0.94), with AUC-ROC approaching 1.00 across all disease classes.
- Integrated dual Explainable AI layers **Grad-CAM** heatmaps and **Partial Layer-wise Relevance Propagation (Partial-LRP)** via attention-matrix rollout demonstrating that Swin's shifted-window mechanism produces

lesion-specific saliency maps whereas ViT diffuses relevance over broader leaf regions.

- Designed deployment-ready inference workflows and model packaging strategies for future edge-device integration.

Applied Machine Learning Research Engineer | Central University of Rajasthan

Jan 2026 – May 2026

- Engineered a biomedical signal-processing pipeline over 11,527 surgical epochs; extracted five EEG sub-bands ($\delta, \theta, \alpha, \beta, \gamma$) via 4-level Discrete Wavelet Transforms (**Sym4**) using **PyWavelets**, with strict zero-phase Notch filtering and artefact-removal boundaries ($BIS \leq 100$).
- Implemented **PyTorch** Categorical GANs (**CAT-GAN**) to synthesise high-fidelity minority-class EEG features, resolving majority-class collapse across non-linear physiological boundaries.
- Deployed **Fuzzy C-Means (FCM)** clustering via **scikit-fuzzy** to map feature centroids, initialising Generalised Bell membership functions to stabilise backpropagation.
- Developed a First-Order Sugeno **ANFIS** engine in **PyTorch** over 400 epochs with **Huber Loss**; achieved **92.19% overall test accuracy** and **92.00% precision** on transitional macro-states; extracted internal parameters (imCO₂/gamma) for XAI verification.

Publications

- **Published Research Paper:** First-author and presenter of peer-reviewed paper “*Asymmetry and Heavy Tails in Bitcoin Volatility: A GARCH Comparative Study*” at the **5th International Conference on Smart Systems: Innovations in Computing**, Manipal University Jaipur (ICSS-2025).

Achievements

- **Secured 2nd Position** out of competitive cohorts in the **Analytix’24** Data Analysis & System Insights Competition, Central University of Rajasthan.

Education

Central University of Rajasthan

2024 – 2026

M.Sc. Computer Science (Specialization: Big Data Analytics) | **CGPA: 8.28/10.0**

- Relevant Coursework: Distributed Data Processing (Hadoop, PySpark), Optimization Techniques, Large Language Models (LLMs), Time-Series Analysis, Data Mining.

University of Delhi

2020 – 2023

B.Sc. Computer Science | **CGPA: 8.56/10.0**

- Core Foundation: Object-Oriented Programming (C++), Data Structures & Algorithms, Database Management Systems (DBMS), Computer System Architecture.